

ALI AKBAR

Dr. Muhammad Shahidullah Hall, University of Dhaka, Ramna, Dhaka-1000, Bangladesh.

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PROFESSIONAL SUMMARY

Experience in RAG systems, LLM agents, and multimodal ML using LangChain, LangGraph, LangSmith, and PyTorch, with 4 research works in ML/computer vision, a UbiComp 2021 runner-up finish, and contribution to an ACL Findings 2025 paper.

EDUCATION

University of Dhaka ([Website](#))

Jan. 2019 - April 2024

Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering

CGPA: 3.52(out of 4.00)

TECHNICAL SKILLS

Languages	C, Java, Python, MATLAB, MySQL, L ^A T _E X.
ML/DL Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV, MLflow, DVC, FastAPI.
LLM & Agentic AI	LangChain, LangGraph, LangSmith, Pydantic, vector DB, Hugging Face, LLM APIs, RAG, Advanced RAG, Prompt Engineering, LoRA, QLoRA.
Data Science Tools	Pandas, NumPy, Matplotlib, MS Excel.
Other Tools	Git, Docker, Streamlit.

EXPERIENCES

Research Assistant | Data & Design Lab, dept. of CSE, University of Dhaka

Feb 2025 – Dec 2025

- Collected real-time field data from substations, developed data analytics solutions, preprocessed and transformed electricity-sector datasets into structured formats for system integration and data-driven decision-making, and contributed to the development of *Biddyut Bandhu*, a domain-specific chatbot.

Volunteer ML Researcher | icddr,b

May 2026 – Present

- Developing a Linux-based shotgun metagenomics pipeline using SRA Toolkit, KneadData, and MetaPhlAn to acquire, preprocess, and generate taxonomic abundance profiles from publicly available healthy-control gut microbiome sequencing data. Integrating these data with organization's existing cohort of 67 IBD patient samples to support comparative microbiome analysis and downstream machine learning studies for identifying disease-associated microbial signatures.

PROJECTS

OCR-Based Handwritten Scanned Document Extraction, Summarization, and Retrieval System.
(Live Demo)

- Developed a RAG-based system for handwritten documents using OCR, semantic retrieval, and FAISS, with source-grounded QA and a HITL workflow for refining automated case summaries.

Conversational AI Chatbot with Memory Using LangGraph and Streamlit.
(Live Demo)

- Developed a multi-turn conversational AI chatbot using LangGraph with graph-based state management, persistent memory, tool calling, and real-time streaming via Groq for context-aware interactions.

Retrieval-Augmented Generation (RAG) System for Multi-Source Website & PDF Question Answering.
(Live Demo)

- Built a web and document-based RAG pipeline that ingests multiple URLs and PDF files, performs semantic retrieval using HuggingFace embeddings and FAISS, and generates source-grounded answers with performance monitoring via LangSmith.

APPLIED ML PROJECTS & RESEARCH

Nurse Care Activity Recognition from Accelerometer Sensor Data Using Fourier- and Wavelet-based Features. (*May, 2021*)

- Engineered and selected time-frequency features from accelerometer data using FFT, DWT, and statistical methods, achieving 87% accuracy with LightGBM for multi-class nurse activity recognition and contributing to a *runner-up (top 2) finish* in the ACM UbiComp 2021 Nurse Care Activity Recognition Challenge.

End-to-End micro-bacterial segmentation and detection using ViT-based hybrid model.

- Developed a MaxViT-based Mask R-CNN for microbacterial instance segmentation, outperforming ResNet50 Mask R-CNN, Mask2Former, and YOLOv11 with 79.31% mAP and 70.01% F1-score, while incorporating ROI Activation CAM for model interpretability (*manuscript will be submitted in an MDPI journal very soon*).

Integrating Gut Microbiome Taxonomic Profiles with Supervised Machine Learning for IBS (Irritable Bowel Syndrome) Diagnosis.

- Built a gut microbiome classification pipeline using 1,109 samples, including newly sequenced Bangladeshi data, achieving 79.73% accuracy and 0.5876 MCC with optimized Logistic Regression for IBS detection.

Thinking Like a Botanist: Challenging Multimodal Language Models with Intent Driven Chain-of-Inquiry

- Contributed to the creation of PlantInquiryVQA, a large-scale plant disease diagnosis dataset with 24,964 images and 138,078 reasoning-based QA pairs for evaluating state-of-the-art multimodal language models.

(*Published in ACL Findings 2025—Acknowledged contributor for large-scale data collection.* [paper link](#))

PUBLICATIONS

Kowshik, M. Ashikuzzaman, Yeasin Arafat Pritom, Md Sohanur Rahman, **Ali Akbar**, and Md Atiqur Rahman Ahad. **”Nurse Care Activity Recognition from Accelerometer Sensor Data Using Fourier-and Wavelet-based Features”** *In Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers*, pp. 434-439. 2021.

ACHIEVEMENTS & AWARDS

- Runner up* at Third Nurse Care Activity Recognition Challenge, 2021, awarded by HASCA workshop, ACM Ubicomp. ([certificate](#))
- Participated* Kharagpur Data Science Hackathon, 2021.

LEADERSHIP & ACTIVITIES

President

Dec 2022 - Feb 2024

Luminator, Dr. Muhammad Shahidullah Hall Students’ Association.

Joint General Secretary

Apr 2023 - Oct 2024

Idrakpur Bus Committee, University of Dhaka.

Organizing Secretary

May 2023 - Aug 2024

Students’ Association of Munshiganj-Bikrampur, University of Dhaka.